

Day 18 (27/11/2025)

Slicing and List Comprehension

```
list1 = [12, 'Python', 87.9, True, 35, 'Java', 88]
```

list 1

```
[12, 'python', 87.9, True, 35, 'Java', 88]
```

~~print~~

Q) list1[0:3] $\Rightarrow$ the range same equal to loops concept

```
[12, 'python', 87.9]
```

Slicing -

[start position : end position : inc/dec position]

default start is 0

default stop position is up to last position

default inc/dec pos is 1

Q) list1[0:-1]

```
[12, 'python', 87.9, True, 35, 'Java']
```

Q) extract last 3 elements

```
list1[-3:]
```

```
[35, 'Java', 88]
```


Q) extract even indexed elements

```
list1 = [12, 'python', 87.9, True, 35, 'Java', 88]
```

0 1 2 3 4 5 6

```
list1[::2]
```

```
[12, 87.9, 35, 88]
```

Q) extract odd indexed elements.

```
list1[1::2]
```

```
['python', True, 'Java']
```

Q) extract elements whose index is multiple of 3

```
list1[3::3]
```

```
[True, 88]
```

Q) reverse the list

```
list1[::-1]
```

```
[88, 'Java', 35, True, 87.9, 'python', 12]
```

Q) write a Program to calculate sum of digits:

```
num = input("enter a value")
```

string method

```
sum  
l = list(num)
```

```
print(l)
```

inp 1278

```
s = 0
```

slp -

```
for i in l:
```

```
    s = s + int(i)
```

['1', '2', '7', '8']

```
print(s)
```

18

Q) write a program to calculate sum of digits.

```
num = int(input("enter a number"))
```

```
s = 0
```

```
while (num > 0):
```

```
    rem = num % 10
```

```
    s = s + rem
```

```
    num = num // 10
```

```
print(s)
```

i/p = 1278

o/p = 18

Q) Remove and add item in a list

write a program to remove the item present at index 4 add it to 2nd position and at the end of the list.

```
sample_list = [34, 54, 67, 89, 11, 43, 94]
```

```
for i in len(sample_list):
```

```
    if i[4] == 4:
```

```
sample_list = [34, 54, 67, 89, 11, 43, 94]
```

```
v = sample_list.pop(4)
```

```
sample_list.insert(2, v)
```

```
sample_list.append(v)
```

```
sample_list
```

```
[34, 54, 11, 67, 89, 43, 94, 11]
```


Q) Concatenate two lists in the following order.

list1 = ["Hello", "take"]

list2 = ["Dear", "Sir"]

Expected result:

['Hello Dear', 'Hello Sir', 'take Dear', 'take Sir']

list1 = ["Hello", "take"]

list2 = ["Dear", "Sir"]

s1 = []

for i in list1:

for j in list2:

s1.append(i+j)

print(s1)

['Hello Dear', 'Hello Sir', 'take Dear', 'take Sir']

o/p

Q) Remove all occurrences of a specific item from a list.

Given a Python list, write a program to remove all occurrences of item 20.

list1 = [5, 20, 15, 20, 25, 50, 20]

Expected output:

[5, 15, 25, 50]


```
list1 = [5, 20, 15, 20, 25, 50, 20]
```

```
nl = []
```

```
for i in list1:
```

```
    if i != 20:
```

```
        nl.append(i)
```

```
print(nl)
```

o/p [5, 15, 25, 50]

Q) Concatenate two lists index-wise

Write a program to add two lists index-wise. Create a new list that contains the 0th index from both the list, then the 1st index item, and so on till the last element, any leftover items will get added at the end of the new list.

```
list1 = ["M", "na", "i", "ke"]
```

```
list2 = ["y", "me", "s", "lly"]
```

```
list1 = ["M", "na", "i", "ke"]
```

```
list2 = ["y", "me", "s", "lly"]
```

```
nl = []
```

```
for i in range(0, 4, 1):
```

```
    s = list1[i] + list2[i]
```

```
    nl.append(s)
```

```
print(nl)
```

o/p ['My', 'name', 'is', 'kelly']